Technical Article

Reduce the Volatility of the IoT with MSP FRAM MCUs

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The expansion of the Internet of Things (IoT) is expected to drive billions of newly connected devices by 2020. One of the primary reasons for connecting these devices is to create more intelligent and automated systems through the processing of data. Any opportunity for these devices to collect more data offers an opportunity to make smarter decisions and improve the overall system.

This is where our microcontrollers with FRAM can add value beyond other solutions in the industry. FRAM, a form of non-volatile RAM, enables faster data storage and nearly unlimited endurance compared to other non-volatile memory technologies. This means that sensing systems can apply mathematical and logical operations to make decisions based on more data in a given period of time. Not only that, the high write endurance equates to extended device longevity as data can continue to be written for many years without degrading the memory. What's more, these advantages for sensor-based systems can be combined to enable system state retention in the event of synchronous or asynchronous power loss for those applications involving regular battery changes or energy harvesting.

Over the past couple of weeks we have showcased the unique values of our microcontroller series with integrated FRAM in a blog series focused on some unique applications:

- How FRAM and CTPL could change stadium lighting
- How FRAM is contributing to portable weather stations
- How FRAM MCUs can achieve higher performance
- How FRAM can help you find your car forever

Intrigued?

Over the past year, TI has released more than 100 device variants in our ultra-low-power FRAM microcontroller family. This portfolio contains offerings from 4-128 KB FRAM and integrated features from ADCs to LCD controllers. Beyond the MCUs themselves, LaunchPad development kits, software and TI Design reference designs are available to make getting started easy! Learn more at www.ti.com/fram.

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